ARIADNE: Grounded and Selective LLM Reasoning for Cross-Camera Trajectory Recovery

Supplementary Material

I. Bootstrap Uncertainty and Voxel Scaling

Bootstrap Precision, Trip-Length Balance, and Voxel Scaling. Table I and Fig. 1 expand on Section IV-B and Table II of the main paper with paired bootstrap intervals and five-split Micro F1 and Vol-MSE comparisons. First, every paired Macro F1 gain is strictly positive across bootstrap resamples, with confidence-interval half-widths narrowing on the larger UrbanVehicle Large split and tightening further against spatial baselines whose differences concentrate on ambiguous junctions. Second, Macro F1 gains exceed Micro F1 gains against VeTrac where sparse side streets offer fewer camera transitions, while matching Micro F1 closely against RETRIEVE across trip lengths. Third, relative 3D Vol-MSE reduction grows from sparse UrbanVehicle, where unobserved mid-block stops limit arrival-time precision, to dense Carla as C2 filters co-clustering detours.

II. Multi-Metric C2 Tiers and Seed Stability

Verification Tiers Across Sparse and Dense Camera Networks. Table II extends Section IV-D of the main paper with the five-metric progression of C2 verification across sparse and dense camera networks. Under sparse UrbanVehicle sensing, Tier 1 shortest-path linking bridges long unmonitored gaps, while Tier 2 speed and span checks filter cross-corridor jumps before C1–C3 turn reasoning. Under dense Carla sensing in plate-free cold start without historical bank entries, pre-stage kinematic DP separates co-clustered trips across junction anchors first. Tier 1–3 zero-detour pruning then removes opposite-lane mid-block snaps on CPU so ARIADNE and Deterministic C2 match. Under synthetic camera-outlier injection in Table III, Deterministic C2 isolates physical bypass behavior from LLM stochasticity. Continuous STLC degrades more gradually than discrete F1 and LCSS under Tier 1+2 because retained anchors bound Euclidean distance across broken spans, while Tier 3's two-hop bypass restores link continuity and reduces multi-seed standard deviation.

III. Multi-Metric Ranker and Vehicle-Level Audit

Length-Weighted Continuity and Per-Vehicle Win/Tie/Loss. Table IV breaks down the ranker comparison in Tables VIII and IX of the main paper by Micro F1, STLC, and per-vehicle Win/Tie/Loss counts against No-Retrieval on Test48 and Val37. On Val37, ungated Top-1 KATE injection lowers STLC below No-Retrieval due to open-boundary truncation, whereas the kinematic gate eliminates all validation losses and attains the highest continuity. On held-out Test48, MLP and GBR regressors trade interior turn wins against boundary losses. The structural gate matches KATE's post-verification scores while abstaining on unanchored boundary clusters because C2 rejects single-source boundary overrides after the call, and the kinematic gate resolves the remaining terminal queue conflicts. Along the Val37 cold-start curve in Fig. 2a, Macro F1 rises and Vol-MSE drops monotonically past KATE by half-bank coverage with zero validation losses, while Fig. 2b confirms flat quality and lower latency across positive routing slacks.

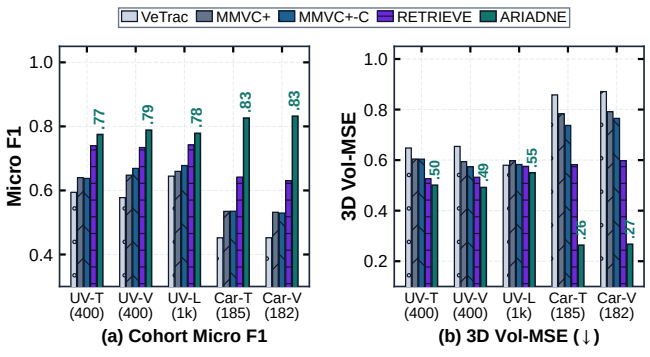


Fig. 1. Benchmark comparison across all five splits on UrbanVehicle and Carla alongside Fig. 5a of the main paper: (a) Length-Weighted Micro F1 and (b) 3D Spatio-Temporal Volume MSE.

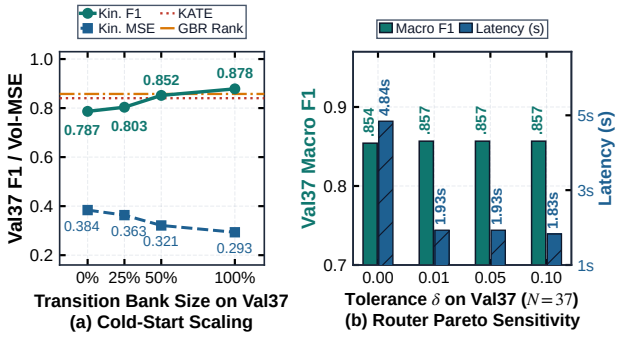


Fig. 2. Cold-start transition-bank scaling and C3 routing slack sensitivity on Val37 alongside Table VIII of the main paper: (a) Macro F1 and Vol-MSE across transition-bank fractions; (b) out-of-sample C3 quality-latency trade-off across routing slack settings.

TABLE I

PAIRED BOOTSTRAP 95% CONFIDENCE INTERVALS AND MULTI-METRIC GAINS OF **ARIADNE (C1–C3)** OVER DOMAIN BASELINES AND UNEXPANDED ANCHORS ACROSS THE THREE URBANVEHICLE SPLITS FROM TABLE II OF THE MAIN PAPER.

Dataset Split	**Baseline / Control**	**Paired ΔMacro F1**	**95% Bootstrap CI**	**CI Half-Width**	**ΔMicro F1**	**ΔSTLC**
UrbanVehicle Test	Nearest-Node Only (0 LLM)	+0.3465	[+0.3214, +0.3713]	±0.0249	+0.3864	+0.1921
($N = 400$)	VeTrac	+0.3014	[+0.2578, +0.3419]	±0.0420	+0.1807	+0.1891
	MMVC+	+0.1490	[+0.1178, +0.1797]	±0.0309	+0.1349	+0.0298
	MMVC+-C	+0.1498	[+0.1169, +0.1817]	±0.0324	+0.1373	+0.0380
	RETRIEVE	+0.0351	[+0.0132, +0.0562]	±0.0215	+0.0351	+0.0241
UrbanVehicle Val	Nearest-Node Only (0 LLM)	+0.3553	[+0.3306, +0.3787]	±0.0240	+0.4006	+0.1970
($N = 400$)	VeTrac	+0.3313	[+0.2878, +0.3741]	±0.0431	+0.2109	+0.2101
	MMVC+	+0.1695	[+0.1395, +0.2008]	±0.0307	+0.1406	+0.0426
	MMVC+-C	+0.1339	[+0.1050, +0.1632]	±0.0291	+0.1196	+0.0254
	RETRIEVE	+0.0550	[+0.0345, +0.0752]	±0.0203	+0.0544	+0.0358
UrbanVehicle Large	Nearest-Node Only (0 LLM)	+0.3880	[+0.3730, +0.4035]	±0.0152	+0.4317	+0.2033
($N = 1{,}000$)	VeTrac	+0.1969	[+0.1722, +0.2214]	±0.0246	+0.1338	+0.1095
	MMVC+	+0.1342	[+0.1143, +0.1531]	±0.0194	+0.1189	+0.0346
	MMVC+-C	+0.1032	[+0.0846, +0.1219]	±0.0186	+0.1007	+0.0128
	RETRIEVE	+0.0323	[+0.0192, +0.0454]	±0.0131	+0.0365	+0.0229

TABLE II

FIVE-METRIC PROGRESSION OF C2 VERIFICATION TIERS ACROSS SPARSE URBANVEHICLE VAL/TEST AND DENSE COLD-START CARLA TEST.

Dataset Split (N)	**Verification Tier / Recovery Configuration**	**Macro F1 ↑**	**Micro F1 ↑**	**STLC ↑**	**LCSS ↑**	**Vol-MSE ↓**
UrbanVehicle Val (400)	w/o C2 (Nearest-Node Anchors Only)	0.4133	0.3878	0.6938	0.3155	0.6638
	C2 Tier 1 Only (Unconstrained Shortest-Path)	0.6332	0.6912	0.8297	0.6356	0.5037
	C2 Tier 1+2 (+ Kinematic Span & Speed Gate)	0.7498	0.7693	0.8844	0.7474	0.5003
	C2 Tier 1–3 (Deterministic C2, 0 LLM Calls)	0.7509	0.7715	0.8852	0.7489	0.4997
	ARIADNE (C1–C3)	**0.7686**	**0.7884**	**0.8908**	**0.7643**	**0.4919**
UrbanVehicle Test (400)	w/o C2 (Nearest-Node Anchors Only)	0.4050	0.3885	0.6872	0.3083	0.6614
	C2 Tier 1 Only (Unconstrained Shortest-Path)	0.6123	0.6676	0.8161	0.6155	0.5195
	C2 Tier 1+2 (+ Kinematic Span & Speed Gate)	0.7238	0.7473	0.8692	0.7230	0.5127
	C2 Tier 1–3 (Deterministic C2, 0 LLM Calls)	0.7258	0.7500	0.8706	0.7248	0.5105
	ARIADNE (C1–C3)	**0.7515**	**0.7749**	**0.8793**	**0.7490**	**0.5010**
Carla Test (185)	w/o C2 (Nearest-Node Anchors Only)	0.4732	0.4976	0.7740	0.5291	0.8462
	DP Pass 1 (Kinematic Sub-Corridor Separation)	0.8069	0.8209	0.8820	0.7374	0.2803
	DP Pass 1+2 (+ Long-Gap Corridor Stitching)	0.8107	0.8242	0.8845	0.7429	0.2776
	ARIADNE (C1–C3) / Deterministic C2 (0 LLM Calls)	**0.8139**	**0.8262**	**0.8919**	**0.7643**	**0.2636**

TABLE III

MULTI-SEED ROBUSTNESS UNDER SYNTHETIC CAMERA-OUTLIER INJECTION ON URBANVEHICLE TEST FROM TABLE V OF THE MAIN PAPER.

Outlier Rate	**C2 Verification Variant**	**Macro F1 ↑**	**Micro F1 ↑**	**STLC ↑**	**LCSS ↑**	**ΔMacro F1 vs. Clean**
5% Outliers	Tier 1 Only	0.5982 ± 0.0033	0.6575 ± 0.0028	0.8078 ± 0.0017	0.5953 ± 0.0035	−1.41 pts
(5 seeds)	Tier 1+2	0.6974 ± 0.0031	0.7239 ± 0.0026	0.8531 ± 0.0021	0.6899 ± 0.0036	−2.64 pts
	Tier 1–3 (Deterministic C2)	$\mathbf{0.7232 \pm 0.0008}$	$\mathbf{0.7475 \pm 0.0006}$	$\mathbf{0.8685 \pm 0.0005}$	$\mathbf{0.7220 \pm 0.0008}$	**−0.26 pts**
10% Outliers	Tier 1 Only	0.5922 ± 0.0071	0.6529 ± 0.0029	0.8011 ± 0.0032	0.5832 ± 0.0067	−2.01 pts
(5 seeds)	Tier 1+2	0.6759 ± 0.0042	0.7049 ± 0.0025	0.8384 ± 0.0021	0.6618 ± 0.0053	−4.79 pts
	Tier 1–3 (Deterministic C2)	$\mathbf{0.7185 \pm 0.0017}$	$\mathbf{0.7429 \pm 0.0014}$	$\mathbf{0.8652 \pm 0.0008}$	$\mathbf{0.7159 \pm 0.0019}$	**−0.73 pts**
20% Outliers	Tier 1 Only	0.5681 ± 0.0080	0.6320 ± 0.0038	0.7856 ± 0.0055	0.5502 ± 0.0098	−4.42 pts
(5 seeds)	Tier 1+2	0.6326 ± 0.0068	0.6650 ± 0.0058	0.8116 ± 0.0046	0.6083 ± 0.0091	−9.12 pts
	Tier 1–3 (Deterministic C2)	$\mathbf{0.7068 \pm 0.0025}$	$\mathbf{0.7309 \pm 0.0030}$	$\mathbf{0.8570 \pm 0.0018}$	$\mathbf{0.7014 \pm 0.0033}$	**−1.90 pts**

TABLE IV

TRAJECTORY CONTINUITY, PER-VEHICLE WIN/TIE/LOSS COUNTS, AND INJECTED CLUSTER COUNTS FOR C1 GATING VS. LEARNED RANKERS.

Evaluation Cohort	**Retrieval / Gating Policy**	**Micro F1 ↑**	**STLC ↑**	**Win / Tie / Loss**	**Injected Clusters**
Held-Out Test48 ($N = 48$)	KATE Cosine Top-1	0.8187	0.9618	16/29/3	48/48 (100.0%)
(No-Retrieval Ref.: 0.7186 Micro F1,	MLP Utility Ranker (ΔF1 Regressor)	0.8063	0.9572	13/33/2	32/48 (66.7%)
0.9244 STLC)	GBR Utility Ranker (ΔF1 Regressor)	0.8166	0.9595	15/30/3	39/48 (81.2%)
	Structural Gate	0.8187	0.9618	16/29/3	31/48 (64.6%)
	Kinematic Gate (Ours)	**0.8285**	**0.9652**	**16/31/1**	**27/48 (56.2%)**
Validation Val37 ($N = 37$)	KATE Cosine Top-1	0.8416	0.9483	12/10/15	37/37 (100.0%)
(No-Retrieval Ref.: 0.7682 Micro F1,	MLP Utility Ranker (ΔF1 Regressor)	0.8295	0.9579	8/22/7	23/37 (62.2%)
0.9574 STLC)	GBR Utility Ranker (ΔF1 Regressor)	0.8501	0.9580	12/17/8	30/37 (81.1%)
	Structural Gate	0.8514	0.9580	11/22/4	24/37 (64.9%)
	Kinematic Gate (Ours)	**0.8609**	**0.9703**	**11/26/0**	**20/37 (54.1%)**